

“Yes/No” Only Please

Let me discuss here the extent of the damage. If you’re dealing with qualitative inference, such as in psychology or medicine, looking for yes/no answers to which magnitudes don’t apply, then you can assume you’re in Mediocristan without serious problems. The impact of the improbable cannot be too large. You have cancer or you don’t, you are pregnant or you are not, et cetera. Degrees of deadness or pregnancy are not relevant (unless you are dealing with epidemics). But if you are dealing with aggregates, where magnitudes do matter, such as income, your wealth, return on a portfolio, or book sales, then you will have a problem and get the wrong distribution if you use the Gaussian, as it does not belong there. One single number can disrupt all your averages; one single loss can eradicate a century of profits. You can no longer say “this is an exception.” The statement “Well, I can lose money” is not informational unless you can attach a quantity to that loss. You can lose all your net worth or you can lose a fraction of your daily income; there is a difference.

This explains why empirical psychology and its insights on human nature, which I presented in the earlier parts of this book, are robust to the mistake of using the bell curve; they are also lucky, since most of their variables allow for the application of conventional Gaussian statistics. When measuring how many people in a sample have a bias, or make a mistake, these studies generally elicit a yes/no type of result. No single observation, by itself, can disrupt their overall findings.

I will next proceed to a sui generis presentation of the bell-curve idea from the ground up.

A (LITERARY) THOUGHT EXPERIMENT ON WHERE THE BELL CURVE COMES FROM

Consider a pinball machine like the one shown in [Figure 8](#). Launch 32 balls, assuming a well-balanced board so that the ball has equal odds of falling right or left at any juncture when hitting a pin. Your expected outcome is that many balls will land in the center columns and that the number of balls will decrease as you move to the columns away from the center.

Next, consider a gedanken, a thought experiment. A man flips a coin and after each toss he takes a step to the left or a step to the right, depending on whether the coin came up heads or tails. This is called the random walk, but it does not necessarily concern itself with walking. You could identically say that instead of taking a step to the left or to the right, you would win or lose \$1 at every turn, and you will keep track of the cumulative amount that you have in your pocket.